
How to Reduce Whisper Hallucination

Husein Zolkepli

Scicom (MSC) Berhad, Malaysia

`husein.zolkepli@scicom.com.my`

Abstract

Whisper is not the most accurate open speech recogniser any more, and it is still the one everybody builds on. It is also the one that writes sentences nobody said. On 42 clips of pure room tone, `whisper-large-v3` emits words on 61.9% of them and emits something on 100% of them. The usual response is to distil a student from Whisper pseudo-labels and filter the hallucinations out of the corpus first. That removes the evidence, not the behaviour: the student is fitted to the subset where the teacher was right, so it inherits a failure mode that never appears in its own training data. The second response, adding non-speech audio so the model learns to stay quiet, is already used in production systems and it works, but it teaches suppression without teaching discrimination. The checkpoint that is best on every non-speech arm here is also the one that recovers the fewest genuinely spoken phrases and deletes 58% of repeated speech. We argue the teacher has to be fixed, with both halves of the signal, and we do the work that requires. We build a benchmark that scores both halves of the problem at once, 11,852 clips over eight arms plus 6,267 synthetic positives, 8,296 clips of real audio that made a production model fail, and FLEURS in 58 languages. We collect a lexicon of 40,891 hallucination phrases in 100 languages, select a text-to-speech system by measurement rather than reputation, and synthesise those phrases as positives so the model sees the same text both as something to suppress and as something to transcribe. We then sweep fine-tuning over rank, learning rate, and the presence of the synthetic positives. The benchmark, the lexicon and the synthetic corpus are released at [Scicom-intl/Whisper-Hallucination](#).

1 Introduction

Whisper [Radford et al., 2022] is not the most accurate open recogniser on the market. It has not been for a while. Canary-1B-v2 [Sekoyan et al., 2025] reports an average word error rate of 8.1% against whisper-large-v3’s 9.9% on its 25 European languages. SeamlessM4T [Seamless Communication team, 2023] reported a 45% relative word error rate reduction over whisper-large-v2 on FLEURS across 77 overlapping languages. Omnilingual ASR [Omnilingual ASR team et al., 2025] covers more than 1,600 languages, over 500 of which had no ASR at all before it.

Whisper is still the model this paper is about, for two reasons.

The first is that one checkpoint covers 99 languages, runs anywhere, and needs no per-language tuning, and that combination is still rare. We measured it on 20 FLEURS [Conneau et al., 2022] test clips in each of 58 languages: whisper-large-v3 reaches a character error rate of 0.075 on the typical language, and 35 of the 58 land under 0.15 (Figure 4).

The second reason is the one that matters. Whisper is the teacher. Distil-Whisper [Gandhi et al., 2023] is trained on Whisper pseudo-labels. So is Kotoba-Whisper [Kotoba Technologies and Ushio, 2024]. So is Granary [Koluguri et al., 2025], and therefore so are Canary-1B-v2 and Parakeet-TDT-0.6B-v3 [Sekoyan et al., 2025], the models that now beat it. A defect in Whisper is not contained to Whisper. It propagates into the corpora that train its successors, and Section 2 argues that the way those corpora are built lets the defect through while deleting the evidence for it.

It also invents text. Not rarely, not only on hard audio, and not harmlessly. Koenecke et al. [2024] ran more than 13,000 clips from AphasiaBank through Whisper and found that roughly 1% of transcriptions contained entire hallucinated phrases or sentences present nowhere in the audio, and that 38% of those hallucinations carried explicit harms: perpetuating violence, inventing associations, or implying false authority. Hallucination rates were higher for speakers with longer non-vocal pauses, which is a symptom of aphasia, so the failure is not distributed evenly across speakers. Barański et al. [2025] come at the same problem from the audio side and show that a curated list of common hallucinations works as a post-processing filter. A recogniser that writes a plausible sentence over silence is worse than one that returns an error, because nothing downstream can tell the difference.

The five checkpoints measured throughout this paper are named as follows. Three are the OpenAI releases. The other two are Mesolitica’s Malay fine-tunes, which matter here because their training pipeline is public and already contains non-speech audio, so they are the closest existing thing to the fix this paper proposes.

Table 1: The checkpoints, and the short labels used in every table and figure below.

label	identifier	what it is
large-v2	openai/whisper-large-v2	base, 1.55 B
large-v3	openai/whisper-large-v3	base, 1.55 B
turbo	openai/whisper-large-v3-turbo	base, 4 decoder layers
malaysian-v2	mesolitica/malaysian-whisper-large-v2	Malay fine-tune of large-v2
malaysian-turbo-v3	mesolitica/Malaysian-whisper-large-v3-turbo-v3	Malay fine-tune of turbo

Table 2: Hallucination on audio that contains no speech. The correct output is the empty string, so any output at all is wrong. The first block counts outputs containing words, the second counts any output including a bare full stop.

arm	n	large-v2	large-v3	turbo	malaysian-v2	malaysian-turbo-v3
silence	42	85.7%	61.9%	59.5%	35.7%	16.7%
music	600	98.7%	97.0%	96.7%	69.7%	31.0%
nonspeech	1,168	98.8%	89.9%	80.7%	76.0%	9.3%
silence any output	42	100%	100%	100%	64.3%	16.7%
music any output	600	100%	100%	100%	73.0%	31.7%
nonspeech any output	1,168	100%	100%	100%	77.8%	9.4%

Every OpenAI checkpoint writes something on every single clip that has no speech in it. Read the first block and the second block together, because published hallucination rates for the same model on the same audio differ by more than ten times depending on which one an author chose to report.

The second failure is repetition. On 1,440 clips built from a known unit repeated a known number of times, the 95th percentile of emitted repeats divided by true repeats is 1.00 for `large-v3` and 18.3 for `turbo`. The two Malay fine-tunes reach 55.0 and 73.3. A single clip can produce hundreds of copies of one token.

Both failures come from the same place. The Whisper decoder is a language model conditioned on an audio encoder. When the acoustic evidence is weak or absent, the prior takes over and the decoder writes fluent text, because writing fluent text is what it was trained to do. Silence is not a token it can emit. It has to decide to stop, and stopping is exactly what it is worst at.

The field’s response has been to work around the model. Put a voice activity detector in front of it, threshold on `no_speech_prob`, block a list of known phrases after the fact. All of these help, and none of them change the weights. Move to a serving stack that does not implement Whisper’s own decoding loop, which is what happens in production, and most of the knobs disappear. vLLM exposes no `no_speech_threshold`, no `compression_ratio_threshold`, no `logprob_threshold`, and no `condition_on_previous_text`. What is left is the model.

2 Distilling from Whisper

The standard way to get a cheaper or a more multilingual Whisper is to run Whisper over unlabelled audio, keep the transcripts, and train a student on them. The pipelines are public and so are the filters, which is what makes this arguable rather than speculative.

Distil-Whisper [Gandhi et al., 2023] pseudo-labels with Whisper and then, in its own words, uses “a simple word error rate heuristic” to select only the highest quality pseudo-labels for training. Kotoba-Whisper [Kotoba Technologies and Ushio, 2024] applies the same recipe to Japanese with `large-v3` as the teacher. NVIDIA’s Granary [Koluguri et al., 2025] builds a 25 language corpus with a pseudo-labeling pipeline whose stages are listed as “segmentation, two-pass inference, hallucination filtering, and punctuation restoration”, and Canary-1B-v2 and Parakeet-TDT-0.6B-v3 [Sekoyan et al., 2025] are trained on it.

Read those two descriptions next to each other. A hallucination filter is a device for removing the teacher’s hallucinations from the student’s training set. It works. The corpus comes out cleaner, and the resulting models are strong. But the examples it deletes are precisely the ones that demonstrate the failure, so the student is fitted to the subset of the teacher’s behaviour that was already correct. It never sees a clip where the teacher wrote `thank you for watching over a music bed`. It has no way to learn that this is wrong. It learns the teacher’s mapping on clean speech and inherits the teacher’s prior everywhere else.

There is a second, quieter effect. Pseudo-label pipelines are built from speech corpora, and speech corpora are segmented to speech. A clip with no voice in it yields a pseudo-label that looks like garbage, so the filter drops it, or it was never in the corpus in the first place. The student therefore has close to zero training examples whose correct output is the empty string. Asking it to say nothing at inference time is asking for behaviour with no support in its training distribution.

We measured how strongly segmentation hides this. GigaSpeech [Chen et al., 2021a] is podcasts and YouTube, which is the right material, and mining it for hallucinations yields 0.41% of clips. AudioSet [Gemmeke et al., 2017], the same kind of source but unsegmented, yields 54.70%. The failures live in what segmentation throws away.

2.1 Negative-only training

The obvious remedy is to put non-speech audio back in. Two groups already do.

Canary-1B-v2 [Sekoyan et al., 2025] states it plainly: the training mix has “non-speech audio added to reduce hallucinations for ASR and AST”. Mesolitica’s Malaysian Whisper fine-tunes [Mesolitica and Malaysia-AI, 2025] do the same thing and, unusually, publish the whole pipeline. Their stage-two

training set includes an AudioSet block with a dedicated noise subset, and the preparation code is open.

It works, and our benchmark shows exactly how well. In Table 2, Malaysian-turbo-v3 is the best checkpoint measured on every non-speech arm, by a wide margin: 16.7% on silence against 61.9% for base large-v3, and 9.3% on non-speech against 89.9%. Adding negatives is not a marginal trick. It is the single most effective thing anyone in this table did.

Then look at what it cost. The same checkpoint recovers 41.1% of phrases that are genuinely spoken, against 69.8% for the base model it was tuned from (Table 6), deletes 58.0% of clips containing repeated speech, and returns character error rate 1.26 on the typical FLEURS language. It did not learn to detect silence. It learned a prior against emitting text, and that prior fires on real speech too.

This is the gap we set out to close. Negative-only supervision teaches suppression without teaching discrimination. The missing ingredient is audio that genuinely contains the phrases the negatives are teaching the model to suppress: a recording where somebody really does say thank you for watching, labelled with that transcript. No public pipeline has one, because nobody collected the phrases first. So the fix has to be applied to the teacher, and it has to supply both halves.

3 Requirements

Three things, in order.

A benchmark that scores both halves. A model that emits nothing is perfect on every non-speech arm and useless. Any metric that looks only at hallucination will select it. The benchmark has to measure suppression and accuracy on the same axis, on the same run.

Knowledge of what the model actually says. Hallucinations are not random. They are a small set of phrases the model falls back on, and they differ by language. That set is enumerable, and once enumerated it can be turned into training data.

Training data where both answers appear. The model needs clips whose correct output is the empty string, and it needs clips where the very same phrases are genuinely spoken and must be transcribed. Without the second half, suppression is learned as a blanket prior against emitting text, which is the failure mode of every aggressive fine-tune we measured.

4 Dataset

4.1 Non-speech audio

Table 3: The eight benchmark arms. Every arm is a test split. The training corpus is a separate build that is disjoint by construction, verified before any training run.

arm	source	clips	correct output
silence	6 noise floors $\times$ 7 durations	42	nothing
music	Free Music Archive [Defferrard et al., 2017]	600	nothing
nonspeech	FSD50K [Fonseca et al., 2020], label-verified voice-free	1,168	nothing
reduplication	a unit repeated an exact number of times	1,440	that many repeats
speech_in_noise	genuine speech mixed at 5 SNRs	1,200	the transcript
genuine	real Malaysian speech	4,694	the transcript
genuine_isolated	one phrase spoken alone	88	the phrase
librispeech_test_clean	LibriSpeech [Panayotov et al., 2015]	2,620	the transcript

The reduplication arm deserves a note, because it is the only arm where both extremes are wrong. A clip contains a unit repeated exactly n times. Emitting more is runaway. Emitting nothing is deletion. A metric that only penalises runaway will reward a model that deletes repeated speech, and one of our own fine-tunes does exactly that.

4.2 Spoken positives

The negative arms ask what a model invents over silence. The positive arm asks what it does when the same phrase is real. This is the half that stops the benchmark from selecting a mute model, and it is built by synthesising the lexicon described in Section 6. It contains 29,112 clips in 83 languages, split into 22,845 train and 6,267 test.

Each test clip is scored three ways, because production audio arrives as a chunk with a noise floor around it, not as a bare 0.9 second clip. The conditions are the clip as published, the clip with two seconds of digital zeros on each side, and the clip with two seconds of real room tone on each side drawn from the `silence` arm. The zeros condition is a control and should do nothing, because Whisper pads every input to a 30 second window anyway.

4.3 Wild audio

Everything above is a built stimulus. The wild pool is real recordings that made a model fail: 3,607 Earnings-22 [Rio et al., 2022] clips carrying human span annotations, 4,689 clips mined from streamed corpora with no annotator, and 187 aphasia clips kept local because the source is membership-gated. It ships as 3,628 train and 4,668 test, split by source recording rather than by clip, because a dozen segments can come from one call and a clip-level split would leak near neighbours.

Mining needs no annotator. Two signatures are self-evident: a token run of six or more, and words emitted where Silero voice activity detection finds no speech. We never call a clip a hallucination because a transcript disagrees with a reference, since that finds ordinary recognition errors.

Table 4: Hallucination yield by corpus, per 8,000 clips heard. Recording quality predicts yield across five thousandfold.

corpus	character	kept	rate
AudioSet [Gemmeke et al., 2017]	YouTube, heavy background noise	4,376	54.70%
AMI [Carletta et al., 2005]	spontaneous meetings, far-field	235	2.94%
GigaSpeech [Chen et al., 2021a]	podcasts and YouTube, speech-aligned	33	0.41%
Earnings-22 [Rio et al., 2022]	conference calls	31	0.39%
People’s Speech <code>dirty</code> [Galvez et al., 2021]	noisy transcripts, clean audio	9	0.11%
VoxPopuli [Wang et al., 2021]	parliament	4	0.05%
People’s Speech <code>clean</code>	curated read speech	1	0.01%

Two lessons. Curated corpora barely fail, so a hallucination study built on them will conclude the problem is small. And a corpus can be the right material and still hide the failure: GigaSpeech is podcasts and YouTube and yields 0.41%, because its segments are cut to speech.

4.4 Multilingual accuracy

`librispeech_test_clean` is English. A multilingual fine-tune can hold its English word error rate and destroy everything else, and nothing in an English benchmark will show it. We add a FLEURS arm: 20 test clips in each of 58 languages, averaged per language rather than per clip so a large language cannot mask a collapsed one, and in character error rate because Chinese, Japanese, Thai, Lao, Burmese and Khmer do not put spaces between words.

This arm paid for itself immediately, as Section 5 shows.

5 Benchmark results

Five checkpoints, greedy decoding, no forced language, no temperature fallback. Every checkpoint is run on all of it, roughly 40,000 clips each.

Three are the OpenAI base models. The other two are Mesolitica’s Malaysian fine-tunes, and they are in the set for a reason: they are the closest public thing to the fix this paper proposes. Their pipeline

is open, their training mix already contains non-speech audio, and they are the only checkpoints here that were deliberately tuned against hallucination. Treat them as the strong baseline, not as a foil.

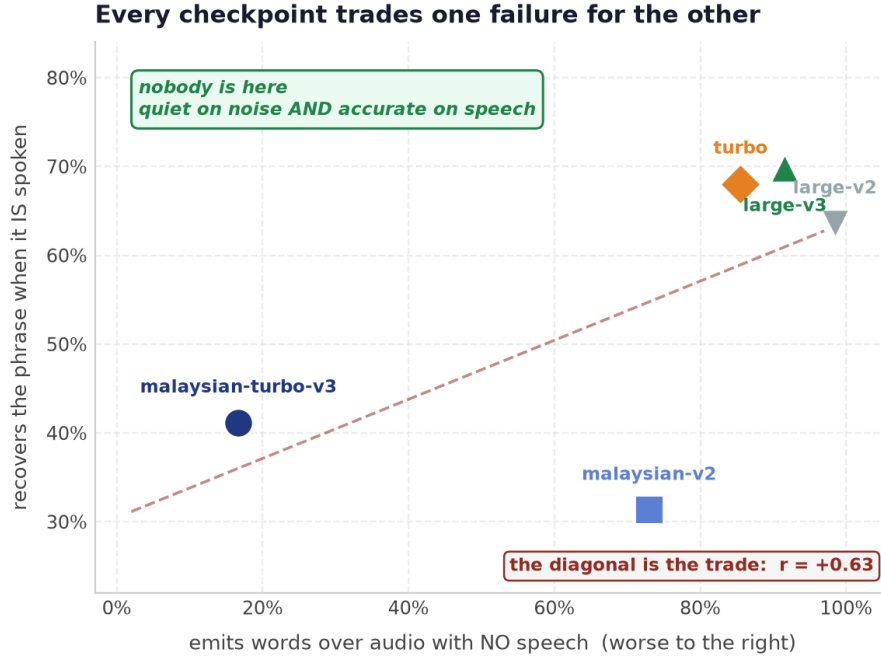


Figure 1: Every checkpoint trades one failure for the other. Horizontal axis is how often the model emits words over audio with no speech in it, pooled over the three non-speech arms and weighted by clip count. Vertical axis is how often it recovers a phrase that is genuinely spoken. The top left corner is the model everyone wants and nothing is there.

No checkpoint is both quiet on noise and accurate on speech. The five sit on a line, $r = +0.63$ between the pooled non-speech word-emission rate and phrase recovery. Quiet models delete real speech and accurate models write over silence. This is the central measurement of the paper and it is what makes a one-sided metric dangerous.

Table 5: Accuracy. The last two rows are the multilingual guard, in character error rate.

arm	n	large-v2	large-v3	turbo	malaysian-v2	malaysian-turbo-v3
librispeech WER	2,620	4.5%	3.5%	3.5%	3.5%	10.6%
genuine WER	4,694	35.4%	31.8%	31.8%	42.3%	60.7%
speech_in_noise WER	1,200	41.6%	34.5%	36.1%	51.1%	58.7%
fleurs CER, typical language	58 langs	0.097	0.075	0.078	1.051	1.259
fleurs CER, macro mean	1,160	0.422	0.305	0.511	1.781	1.681

English accuracy says nothing about the other 57 languages. malaysian-whisper-v2 matches base large-v3 on LibriSpeech, 3.5% either way, and returns character error rate 1.05 on the typical FLEURS language. Two of its 58 languages land under 0.15, against 35 for large-v3. Specialising on Malay cost everything else, and no English benchmark can see it.

The macro mean and the median disagree by four times because Whisper cannot transcribe some of these languages at all. For base large-v3, Amharic sits at 1.81 character error rate, Khmer at 1.58, Somali at 1.44, while Spanish, Italian and Dutch sit at 0.01. The mean measures the tail and the median measures the typical language. Quoting one without the other misleads.

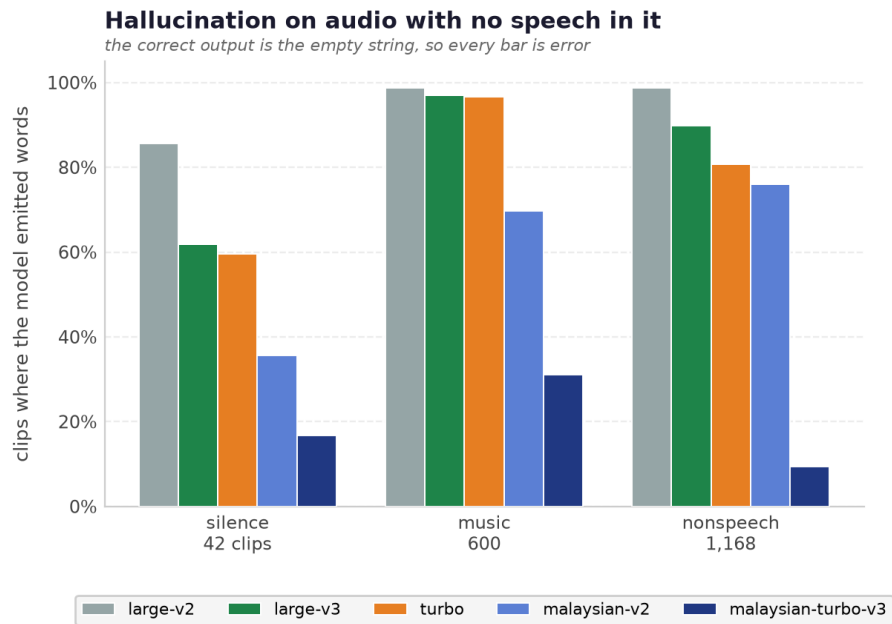


Figure 2: Hallucination on the three non-speech arms. Every bar is error, because the correct output is the empty string.

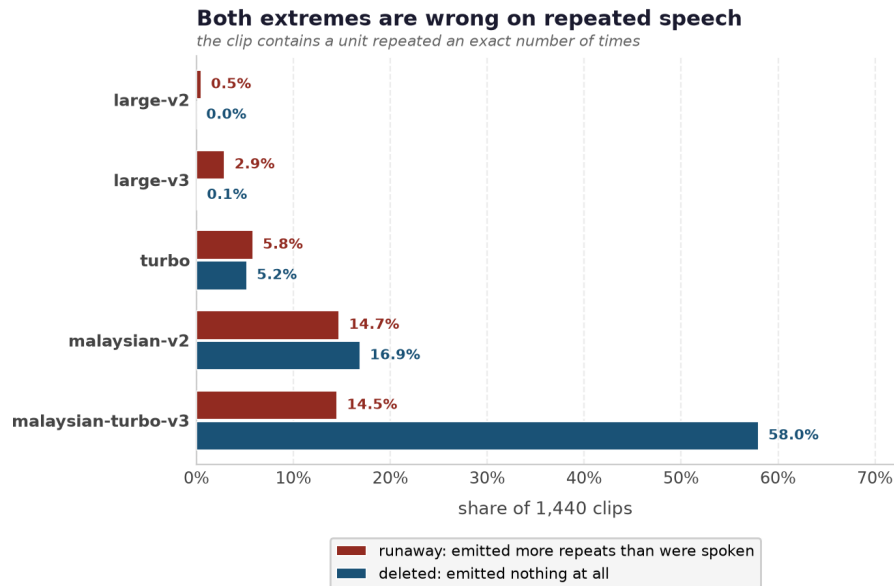


Figure 3: Both extremes are wrong on repeated speech. A model that emits more repeats than were spoken is running away; a model that emits nothing has deleted the speech. Malaysian-turbo-v3 does the second on 58.0% of clips.

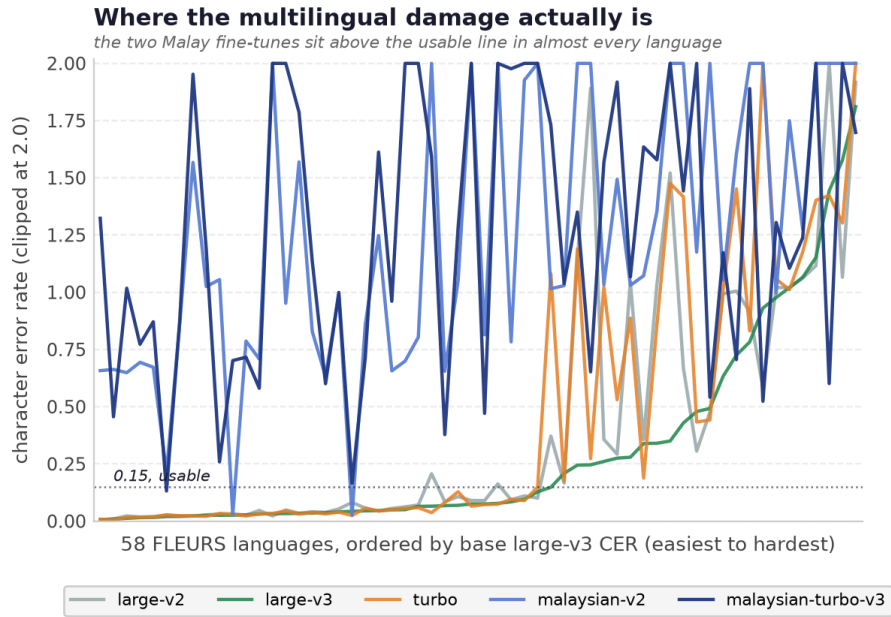


Figure 4: The same measurement language by language, ordered by base large-v3 difficulty. Whisper genuinely cannot transcribe the right hand end: Amharic, Khmer and Somali sit above 1.4 for every checkpoint. The two Malay fine-tunes sit above the usable line almost everywhere, which is the damage LibriSpeech cannot report.

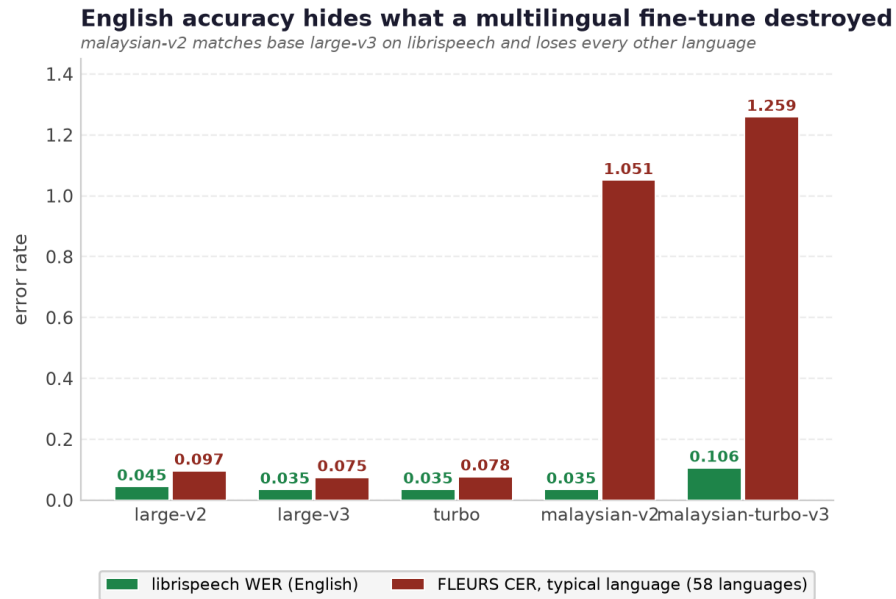


Figure 5: English accuracy hides what a multilingual fine-tune destroyed. malaysian-whisper-v2 matches base large-v3 on LibriSpeech and returns character error rate 1.05 on the typical FLEURS language.

malaysian-turbo-v3 looks broken in Table 5 and is not. Drop the roughly 1% of clips carrying a token run of six or more and genuine falls from 60.7% to 33.9%, and speech_in_noise from 58.7% to 25.6%, the best in the table. One percent of clips carries 27 points of word error rate. Looping is also what separates turbo from large-v3 on FLEURS, 4.7% of clips against 0.9%.

5.1 The positive arm

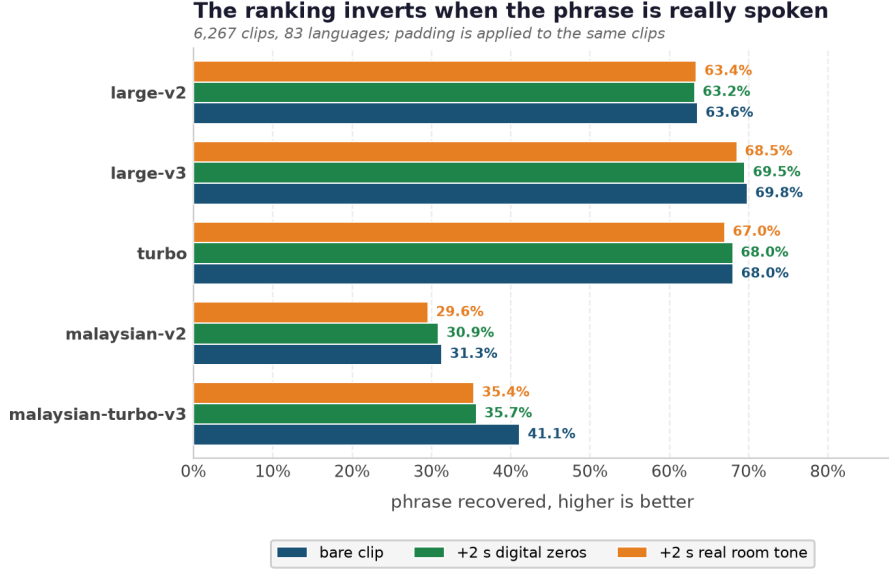


Figure 6: The ranking inverts when the phrase is really spoken. 6,267 clips in 83 languages, each run three ways. Padding is applied to the same clips, so each triple is a within-clip comparison.

Table 6: lexicon_synth test split. Emitting the phrase is correct here.

model	recovered (bare to room tone)	median CER	mean CER	> 2× output
whisper-large-v2	63.6% to 63.4%	0.125	0.378 to 0.384	0.6% to 0.7%
whisper-large-v3	69.8% to 68.5%	0.089	0.288 to 0.302	0.2% to 0.4%
whisper-large-v3-turbo	68.0% to 67.0%	0.087	0.348 to 0.382	0.7% to 0.9%
malaysian-whisper-v2	31.3% to 29.6%	0.800	0.770 to 1.151	2.6% to 3.7%
Malaysian-turbo-v3	41.1% to 35.4%	0.385	1.508 to 3.749	3.0% to 5.7%

Malaysian-turbo-v3 wins every non-speech arm in Table 2 and recovers 41.1% of genuinely spoken phrases, against 69.8% for large-v3. Its silence is a prior against emitting text, not a detector. Report one side only and you will ship it.

Room tone costs the fine-tunes and not the base models. Same speech, two seconds of noise floor on each side: malaysian-v2 gains 0.38 mean character error rate, malaysian-turbo-v3 gains 2.24, large-v3 gains 0.014. Mean character error rate here is a tail statistic. The median moves by hundredths while the mean triples, because 3% to 6% of clips run past twice the phrase length.

5.2 Results on wild audio

On 4,421 clips that a voice activity detector confirms contain no speech, every OpenAI checkpoint emits words on 100% of them. Malaysian-turbo-v3 emits words on 8.5%. Of those outputs, 53.6% for large-v3 and 72.1% for turbo are exactly a phrase from the lexicon, against roughly

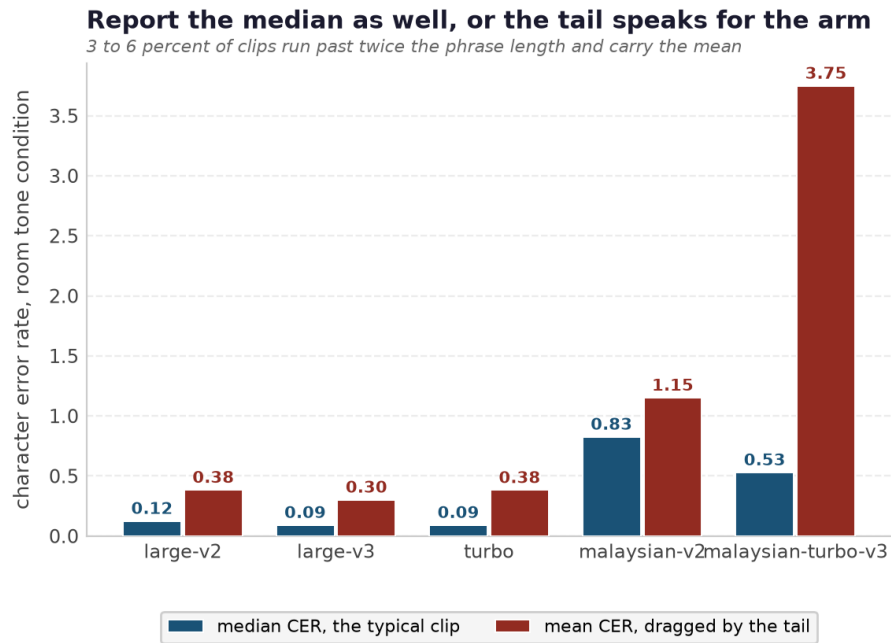


Figure 7: Mean character error rate on the positive arm is a tail statistic. The median barely moves between checkpoints while the mean triples, because a few percent of clips run past twice the phrase length. Report both.

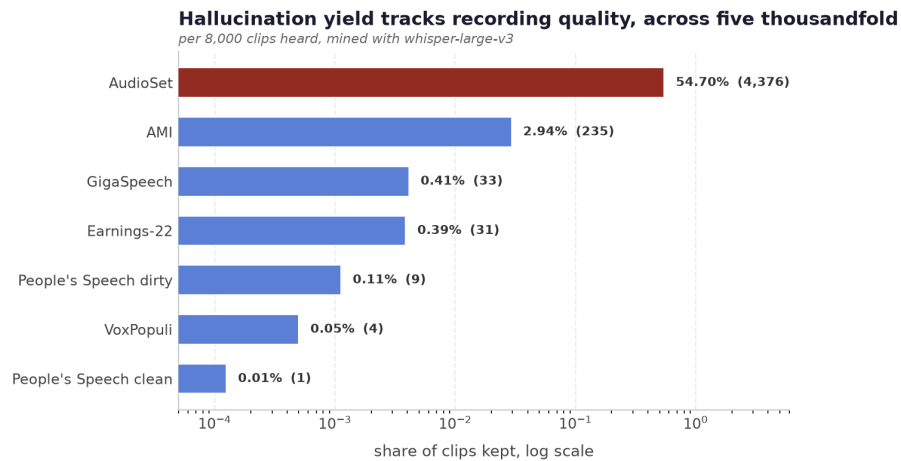


Figure 8: Hallucination yield by corpus, per 8,000 clips heard, on a log scale. Curated corpora barely fail. GigaSpeech is the right material and still yields 0.41%, because its segments are cut to speech.

18% to 22% on speech clips from the same corpus. The blocklist works, but only once the audio is already known to be blank.

Most wild loops are model-specific. On clips mined because large-v3 looped, turbo loops on 17.2%. A loop corpus mined with one model is not a fair loop benchmark for another.

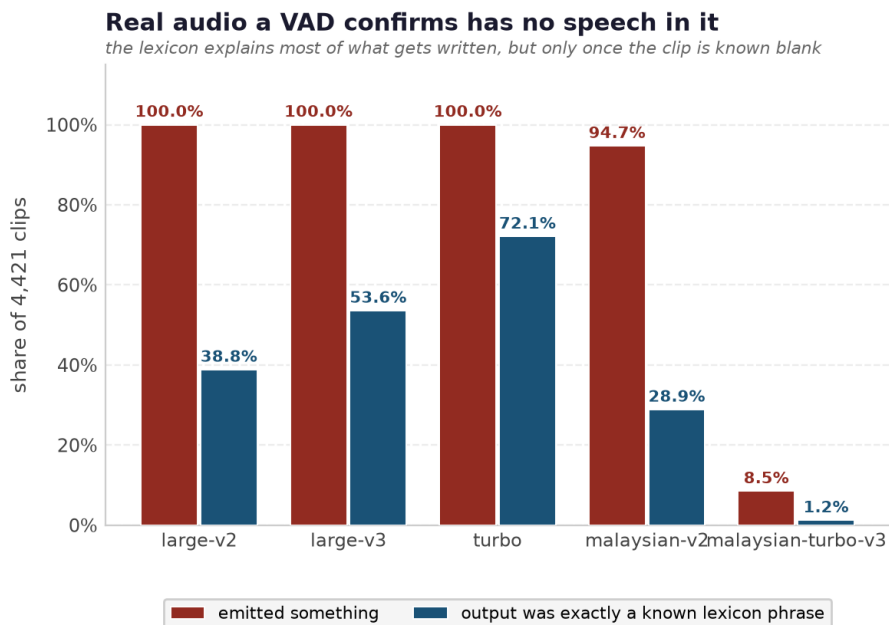


Figure 9: Real audio that a voice activity detector confirms has no speech in it. Every OpenAI checkpoint writes something on all 4,421 clips. Most of what they write is a phrase from the lexicon, which is what makes a blocklist work, but only once the clip is already known to be blank.

6 Lexicon

Hallucinations repeat themselves. thank you for watching, so, the Russian *Prodolzhenie sleduet...* (“to be continued”), and the Malay *terima kasih kerana menonton* appear over and over on audio containing none of them. We merged four public sources with our own mining into 40,891 phrases across 100 languages.

The lexicon is the bridge between the two halves of this work. Read one way it is a blocklist. Read the other way it is a list of sentences the model must still be able to transcribe when somebody genuinely says them, which is exactly the training signal that is missing.

It needs filtering before use. It contains Whisper’s own loop artefacts captured as phrases. One entry is a single Gujarati character repeated seven times. These wreck character error rate comparisons, one of them produced a CER of 22.9. We drop any entry that is a single repeated token or has two or fewer distinct characters.

It also needs a frequency cut. Of 39,476 queued entries, 35,594 were observed exactly once, and 27,974 of those run five words or longer. They are raw recognition fragments, not phrases: 0073a this vehicle s got 72 00 miles on it 3 5l v engine. Synthesising those is pointless, because the round trip fails by construction on digits and truncation. Two independent text-to-speech systems score character error rate near 1.0 on them, which is the signal that the input is at fault rather than the generator. We synthesise entries observed at least twice: 3,882 entries across 85 languages, median two words.

7 Synthesis

To turn the lexicon into training data we need each phrase spoken aloud, in its own language, in more than one voice. That is a text-to-speech problem, and the choice of system matters more than it looks.

7.1 Choosing a generator

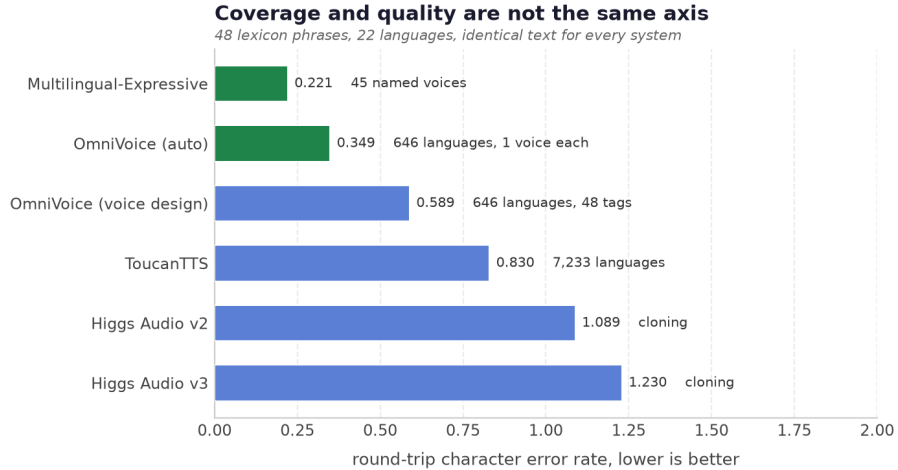


Figure 10: Text-to-speech candidates on 48 lexicon phrases in 22 languages, identical text, scored by recognition round trip. Coverage and quality are not the same axis.

Table 7: Text-to-speech candidates. Round-trip character error rate, lower is better.

	mean CER	median	wins	languages	voices
Multilingual-Expressive-TTS-1.7B	0.221	0.000	16	untagged	45 named
OmniVoice (auto)	0.349	0.018	2	646	1 per language
OmniVoice (voice design)	0.589	0.062	3	646	48 tags
ToucanTTS	0.830	0.586	0	7,233	sampled from a GAN
Higgs Audio v2	1.089	0.713	0	untagged	cloning
Higgs Audio v3	1.230	0.483	1	82/100	cloning

Coverage and quality are not the same axis. ToucanTTS covers 7,233 languages and wins nothing. OmniVoice covers 646 and is the only system whose coverage is enumerable, so it drives the hundred-language sweep, with Multilingual-Expressive handling Malay, English, Chinese and Tamil.

7.2 Voice diversity

OmniVoice takes no speaker argument, so every clip it produces carries an empty voice field. We measured what that means with WavLM speaker embeddings [Chen et al., 2021b] on a calibrated scale: 33 of its 74 languages sat at a median pairwise cosine of 0.75 or worse. One voice per language. A corpus built that way teaches the model one speaker’s phrase, not the phrase.

The obvious fix, reference cloning, is the wrong one.

Cloning costs about half the yield and separates the voices less well than simply naming a speaker. So we topped up 26 languages rather than replacing them: three named voices per phrase, auto-mode clips kept. 12,190 of 15,675 new clips passed the quality gate, 77.8%, and the median collapsed language moved from 0.81 to 0.63.

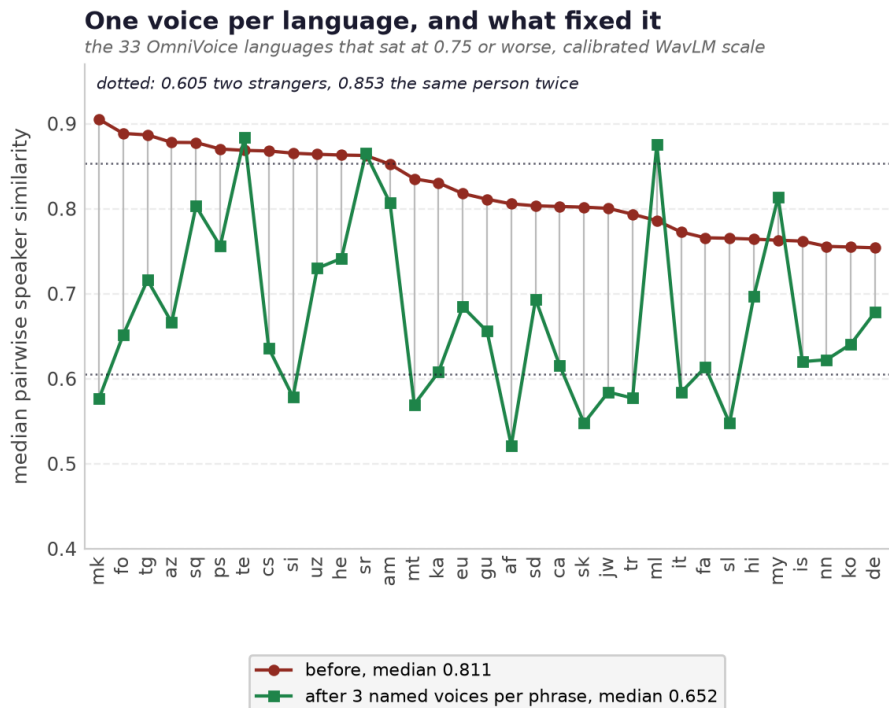


Figure 11: One voice per language, and what fixed it. Pairwise speaker similarity on a calibrated scale where 0.605 is two strangers and 0.853 is two clips of the same person.

Table 8: Two routes to distinct voices, probed on the same 4 languages, 24 phrases and 4 voices. Lower cosine means better separated voices.

route	yield mk / gu / it	median CER	different-voice cosine
Multilingual-Expressive, speaker name	86% / 80% / 79%	0.07 / 0.17 / 0.00	0.65 / 0.64 / 0.58
OmniVoice, reference cloning	46% / 68% / 53%	0.32 / 0.25 / 0.26	0.68 / 0.72 / 0.69
auto mode, for reference	89% / 73% / 83%	0.02 / 0.11 / 0.00	0.91 / 0.81 / 0.77

7.3 Voice conversion

Character error rate alone picks the wrong winner here, because a converter that returns its input unchanged scores a perfect zero degradation and is worthless. We report the cost added on top of the source clip, similarity to the target and to the source on the same calibrated scale, and the gap between them, which is the conversion that actually happened.

7.4 Quality gate

Every synthesised clip passes a recognition round trip: the recogniser must recover the phrase, and the clip must run about as long as the phrase should take. The gate is per language and anchored to the judge’s own floor, which is Whisper’s character error rate on real FLEURS speech in that language. A language where the judge itself scores 0.88, as it does on Burmese, cannot be held to a 0.25 gate. Nine such languages are excluded rather than scored.

The final corpus is 29,112 clips in 83 languages, 22,845 train and 6,267 test, 15.8 hours, 92% non-English.

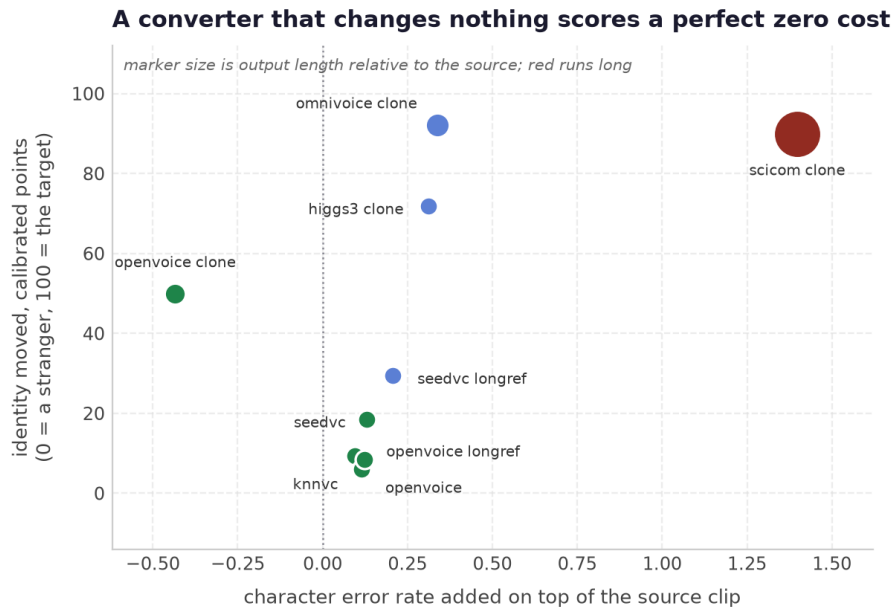


Figure 12: Voice conversion candidates. Character error rate added on top of the source clip against how far the voice actually moved. A converter that returns its input scores a perfect zero cost and is worthless, which is why the vertical axis is here at all.

8 Training sweep

8.1 Setup

We fine-tune whisper-large-v3 with LoRA [Hu et al., 2021] and with full fine-tuning. The target for a blank clip is the empty transcript under that clip’s language token, which is what teaches “no words here” rather than “emit something short”. The four prompt tokens are masked out of the loss, so loss falls only on the transcript and the final end-of-text token.

Table 9: Training sets. Everything is disjoint from the benchmark by construction and every test split stays held out.

source	clips	hours	target	teaches
corpus nonspeech (FSD50K dev)	16,983	33.3	empty	noise means say nothing
corpus music (FMA shards 2-12)	7,170	58.3	empty	music means say nothing
corpus Malay speech (Emilia [He et al., 2024])	5,801	14.8	transcript	keep accuracy
corpus reduplication	2,514	4.0	exact repeats	do not run away
corpus silence	59	0.7	empty	room tone means say nothing
lexicon_synth train	22,845	12.5	the phrase	the phrase is real, transcribe it
wild train	3,656	9.6	empty	real noisy audio means say nothing

8.2 Mixes

The sweep answers a paired question, and both halves are run at every configuration.

1. Without the synthetic lexicon, does fine-tuning improve hallucination and repetition on real audio?
2. With the synthetic lexicon, does it improve them further, and what does it cost?

Two mixes, identical except for the synthetic positives.

Table 10: The paired mixes. `no_synth` is all with `lexicon_synth` removed and nothing else changed.

mix	clips	blank share	contains <code>lexicon_synth</code>
<code>no_synth</code>	36,183	77%	no
<code>all</code>	59,028	47%	yes, 22,845 clips

Removing the synthetic positives also moves the blank share from 47% to 77%. That is not a confound to be stripped out, it is the mechanism. The synthetic positives are the counterweight, and removing them necessarily makes the mix more blank-dominated.

8.3 Grid

Twelve configurations per mix, twenty-four runs in total. LoRA alpha tracks twice the rank so rank is the only thing that varies. Adapters cover the attention projections and the two feed-forward layers, because those hold most of the parameters in a Whisper block. Full fine-tunes use batch 4 with 4 accumulation steps to hold the same 16 clips per optimiser step as LoRA’s 8 with 2, so effective batch size does not confound the comparison.

Table 11: The grid. `1e-3` is absent on purpose, see below.

method	rank / alpha	trainable params	learning rates
LoRA	32 / 64	57.7 M (3.6%)	1e-4, 2e-4, 5e-4
LoRA	64 / 128	115.3 M (7.0%)	1e-4, 2e-4, 5e-4
LoRA	128 / 256	230.7 M (13.0%)	1e-4, 2e-4, 5e-4
full fine-tune		1,574.9 M (100%)	5e-6, 1e-5, 2e-5

Every run is trained for 1,000 steps with 50 warmup steps in bf16, then scored on the complete benchmark: the eight published arms, the synthetic positives, the wild test split, and FLEURS.

Control the optimiser before attributing anything to the data. Our first sweep ran at `1e-3`. The corpus mix finished at loss 9.76 while mixes containing the same clips finished at 1.1 to 1.2, which reads as evidence that a blank-heavy mix is bad. It is not. The same mix at `2e-4` converges cleanly to 0.64. The checkpoint from the diverged run hallucinates on 100% of silence and 100% of music, which is a broken optimiser, not a data result.

8.4 Results

9 Limitations

The positive pool is largely synthetic. It is 52% synthetic text-to-speech and 23% read prose, and only 25% is real conversational speech, which is Sarawak dialect. A model tuned on synthesised phrases may be learning something about the generator as well as about the phrases.

There is no telephony condition. Everything is 16 kHz, and a great deal of production audio is 8 kHz narrowband.

Chinese and Tamil positives are close to absent, 11 and 4 clips.

The `silence` arm is 42 clips, so its resolution is 2.4 percentage points per clip. Differences of a few points on that arm are noise. The `music` arm may contain singing, since its source labels vocals as unknown. `nonspeech` is the label-verified alternative and should be preferred when the two disagree.

`speech_in_noise` is synthetic mixing, so the signal-to-noise ratio is exact but there is no Lombard effect and no channel.

The synthetic corpus was filtered using `whisper-large-v3` with the language forced, so that checkpoint is scored partly on clips it already agreed with. This flatters it against `large-v2` and `turbo`. It does not explain a thirty point gap to the Malay fine-tunes.

10 Future work

The decoder-side ablation is designed and not run: 34 configurations of beam width, temperature fallback, repetition and presence penalties, and the interaction between those and a fine-tuned checkpoint. The interesting question is whether a model trained to stop still needs the knobs.

Stage two of the sweep is the winning configuration applied to `whisper-large-v3-turbo`, which is the checkpoint most people actually serve and the one with the worst repetition profile of the three base models.

Two languages, Sinhala and Malagasy, have no FLEURS configuration, so they have no judge floor and fall back to a flat gate that accepts almost nothing. They need a floor measured from another corpus before their phrases can be synthesised reliably.

Finally, the wild pool should grow. AudioSet yields more than half of what it is asked for, so the bottleneck is not availability, it is compute and storage. A wild benchmark an order of magnitude larger would let hallucination be measured per acoustic condition rather than in aggregate.

11 Acknowledgement

This work was carried out at Scicom (MSC) Berhad. The author thanks Scicom for the compute and for the time to publish the negative results alongside the positive ones.

References

- Mateusz Barański, Jan Jasiński, Julitta Bartolewska, Stanisław Kacprzak, Marcin Witkowski, and Konrad Kowalczyk. Investigation of whisper asr hallucinations induced by non-speech audio, 2025. URL <https://arxiv.org/abs/2501.11378>.
- Jean Carletta, Simone Ashby, Sebastien Bourban, Mike Flynn, Mael Guillemot, Thomas Hain, Jaroslav Kadlec, Vasilis Karaiskos, Wessel Kraaij, Melissa Kronenthal, Guillaume Lathoud, Mike Lincoln, Agnes Lisowska, Iain McCowan, Wilfried Post, Dennis Reidsma, and Pierre Wellner. The ami meeting corpus: A pre-announcement. In *International Workshop on Machine Learning for Multimodal Interaction*, 2005.
- Guoguo Chen, Shuzhou Chai, Guanbo Wang, Jiayu Du, Wei-Qiang Zhang, Chao Weng, Dan Su, Daniel Povey, Jan Trmal, Junbo Zhang, Mingjie Jin, Sanjeev Khudanpur, Shinji Watanabe, Shuaijiang Zhao, Wei Zou, Xiangang Li, Xuchen Yao, Yongqing Wang, Yujun Wang, Zhao You, and Zhiyong Yan. Gigaspeech: An evolving, multi-domain asr corpus with 10,000 hours of transcribed audio, 2021a. URL <https://arxiv.org/abs/2106.06909>.
- Sanyuan Chen, Chengyi Wang, Zhengyang Chen, Yu Wu, Shujie Liu, Zhuo Chen, Jinyu Li, Naoyuki Kanda, Takuya Yoshioka, Xiong Xiao, Jian Wu, Long Zhou, Shuo Ren, Yanmin Qian, Yao Qian, Jian Wu, Michael Zeng, and Furu Wei. Wavlm: Large-scale self-supervised pre-training for full stack speech processing, 2021b. URL <https://arxiv.org/abs/2110.13900>.
- Alexis Conneau, Min Ma, Simran Khanuja, Yu Zhang, Vera Axelrod, Siddharth Dalmia, Jason Riesa, Clara Rivera, and Ankur Bapna. Fleurs: Few-shot learning evaluation of universal representations of speech, 2022. URL <https://arxiv.org/abs/2205.12446>.
- Michael Defferrard, Kirell Benzi, Pierre Vandergheynst, and Xavier Bresson. Fma: A dataset for music analysis, 2017. URL <https://arxiv.org/abs/1612.01840>.
- Eduardo Fonseca, Xavier Favory, Jordi Pons, Frederic Font, and Xavier Serra. Fsd50k: An open dataset of human-labeled sound events, 2020. URL <https://arxiv.org/abs/2010.00475>.
- Daniel Galvez, Greg Diamos, Juan Ciro, Juan Felipe Ceron, Keith Achorn, Anjali Gopi, David Kanter, Maximilian Lam, Mark Mazumder, and Vijay Janapa Reddi. The people’s speech: A large-scale diverse english speech recognition dataset for commercial usage, 2021. URL <https://arxiv.org/abs/2111.09344>.
- Sanchit Gandhi, Patrick von Platen, and Alexander M. Rush. Distil-whisper: Robust knowledge distillation via large-scale pseudo labelling, 2023. URL <https://arxiv.org/abs/2311.00430>.
- Jort F. Gemmeke, Daniel P. W. Ellis, Dylan Freedman, Aren Jansen, Wade Lawrence, R. Channing Moore, Manoj Plakal, and Marvin Ritter. Audio set: An ontology and human-labeled dataset

- for audio events. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2017.
- Haorui He, Zengqiang Shang, Chaoren Wang, Xuyuan Li, Yicheng Gu, Hua Hua, Liwei Liu, Chen Yang, Jiaqi Li, Peiyang Shi, Yuancheng Wang, Kai Chen, Pengyuan Zhang, and Zhizheng Wu. Emilia: An extensive, multilingual, and diverse speech dataset for large-scale speech generation, 2024. URL <https://arxiv.org/abs/2407.05361>.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models, 2021. URL <https://arxiv.org/abs/2106.09685>.
- Allison Koenecke, Anna Seo Gyeong Choi, Katelyn X. Mei, Hilke Schellmann, and Mona Sloane. Careless whisper: Speech-to-text hallucination harms. In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 2024. URL <https://arxiv.org/abs/2402.08021>. arXiv:2402.08021.
- Nithin Rao Koluguri, Monica Sekoyan, George Zelenfroynd, Sasha Meister, Shuoyang Ding, Sofia Kostandian, He Huang, Nikolay Karpov, Jagadeesh Balam, Vitaly Lavrukhin, Yifan Peng, Sara Papi, Marco Gaido, Alessio Brutti, and Boris Ginsburg. Granary: Speech recognition and translation dataset in 25 european languages. In *Interspeech*, 2025. URL <https://arxiv.org/abs/2505.13404>. arXiv:2505.13404.
- Kotoba Technologies and Asahi Ushio. Kotoba-whisper: Distilled whisper for japanese automatic speech recognition. Hugging Face model card, 2024. URL <https://huggingface.co/kotoba-tech/kotoba-whisper-v2.0>.
- Mesolitica and Malaysia-AI. Malaysian whisper: Open speech recognition fine-tunes and training pipeline for malaysian languages. Hugging Face model and dataset cards; pipeline at <https://github.com/malaysia-ai/dataset/tree/main/stt-whisper>, 2025. URL <https://huggingface.co/mesolitica/Malaysian-whisper-large-v3-turbo-v3>.
- Omnilingual ASR team, Gil Keren, et al. Omnilingual asr: Open-source multilingual speech recognition for 1600+ languages, 2025. URL <https://arxiv.org/abs/2511.09690>.
- Vassil Panayotov, Guoguo Chen, Daniel Povey, and Sanjeev Khudanpur. Librispeech: An asr corpus based on public domain audio books. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2015.
- Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision, 2022. URL <https://arxiv.org/abs/2212.04356>.
- Miguel Del Rio, Peter Ha, Quinten McNamara, Corey Miller, and Shipra Chandra. Earnings-22: A practical benchmark for accents in the wild, 2022. URL <https://arxiv.org/abs/2203.15591>.
- Seamless Communication team. Seamlessm4t: Massively multilingual & multimodal machine translation, 2023. URL <https://arxiv.org/abs/2308.11596>.
- Monica Sekoyan, Nithin Rao Koluguri, Nune Tadevosyan, Piotr Zelasko, Travis Bartley, Nick Karpov, Jagadeesh Balam, and Boris Ginsburg. Canary-1b-v2 & parakeet-tdt-0.6b-v3: Efficient and high-performance models for multilingual asr and ast, 2025. URL <https://arxiv.org/abs/2509.14128>.
- Changhan Wang, Morgane Riviere, Ann Lee, Anne Wu, Chaitanya Talnikar, Daniel Haziza, Mary Williamson, Juan Pino, and Emmanuel Dupoux. Voxpopuli: A large-scale multilingual speech corpus for representation learning, semi-supervised learning and interpretation, 2021. URL <https://arxiv.org/abs/2101.00390>.